MUHAMMAD ZAID TAHIR

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PROFESSIONAL SUMMARY

Emerging AI/ML Engineer with hands-on expertise in Generative AI, LLM orchestration, LangChain, LangGraph, RAG pipelines, and Agentic AI systems. Skilled in building end-to-end AI-powered full-stack applications using Python, FastAPI, Flask, and React, with a strong foundation in machine learning, deep learning, NLP, and computer vision. Proven track record in international hackathons and competitions including Harvard CS50x Puzzle Day (9/9 and 10/10 perfect scores) and UC Berkeley CALICO Informatics Competition (Global Rank 292/820). Currently deepening expertise in production-grade Agentic AI, vector databases, and LLM fine-tuning.

CORE COMPETENCIES

- Machine Learning & AI: Classification, regression, ensemble methods, KNN, Decision Trees, Random Forest, SVC, Gradient Boosting.
- Data Preprocessing & Feature Engineering: Handling missing data, SMOTE, scaling, one-hot encoding, GridSearchCV, cross-validation.
- Python Programming: NumPy, Pandas, scikit-learn, Matplotlib, Seaborn for data analysis and visualization.
- AI Application Development: Flask, Pickle model deployment, Streamlit, REST APIs for real-time ML applications.
- Recommendation Systems: Collaborative filtering, content-based filtering, hybrid models, TF-IDF vectorization.

EDUCATION

BS Computer Science — **09/2023 - 17/2027**
UAF, Faisalabad

- Computer Science
- CGPA: 3.3

Intermediate — **04/2021 - 06/2013**
KIPS College

- FSC

INTERNSHIP

Career Institute — **06/2025 - 08/2025**

Completed a structured AI/ML internship program, gaining hands-on experience in building and evaluating machine learning models for classification and prediction tasks.

- Applied data preprocessing, feature engineering, and model evaluation techniques using Python, scikit-learn, pandas, and NumPy.
- Developed small projects/prototypes to implement end-to-end ML pipelines, including model training, testing, and deployment simulations.
- Learned best practices in ML workflow, version control, and reporting results, preparing for real-world AI/ML applications

PERSONAL PROJECTS

. AI Communication Coach

- Designed and developed a multimodal AI-powered platform that analyzes speech, vocal tone, and body language to provide actionable communication feedback.
- Implemented speech analysis (transcription with Whisper, filler word detection, pace measurement, sentiment detection) and visual analysis (eye contact, posture, gestures, and stability tracking) to deliver holistic insights.
- Built a full-stack solution with Flask, SocketIO, and WebRTC for real-time coaching, integrating Whisper (speech recognition), MediaPipe (visual tracking), HuggingFace Transformers (sentiment analysis), and SpeechBrain (emotion recognition).

- Delivered real-time coaching and detailed PDF feedback reports, enabling measurable improvements in presentation skills and communication effectiveness.

Breast Cancer Classification (Machine Learning Project)

- Built a predictive model to classify breast cancer tumors as benign or malignant using structured diagnostic data.
- Designed a robust ML pipeline integrating SMOTE for class imbalance correction, feature scaling, and K-Nearest Neighbors (KNN) with hyperparameter optimization.
- Applied GridSearchCV with Stratified K-Fold cross-validation to fine-tune model parameters, achieving ~97% accuracy with precision and recall above 95%.
- Validated performance through a confusion matrix and classification report, ensuring model reliability for real-world clinical use cases.

MediBot – AI-Powered Medical Assistant (Neura Hackathon 2025)

- Designed and developed an AI medical assistant web app to analyze user symptoms, suggest potential conditions, and provide Pakistan-specific medicine and hospital recommendations.
- Implemented a responsive frontend with real-time chat and auto-resizing inputs, connected to a Flask + SQLite backend for smooth, scalable functionality.
- Integrated Google Gemini for conversational AI and REST APIs for hospital and medicine lookup, enabling dynamic, real-time recommendations.
- Delivered a fully functional demo within the hackathon timeframe, demonstrating practical impact on accessible healthcare for underserved communities.

HackNation Global AI Hackathon 2025 (in collaboration with MIT Sloan AI Club)

- Developed an end-to-end AI system for real-time pricing optimization, integrating competitor analysis and multiple strategies (Conservative, Standard, Aggressive) to maximize profitability.
- Built a Flask backend with a trained ML model and responsive web frontend for seamless, real-time pricing recommendations.
- Trained the model on 676 retail pricing records across 9 product categories, achieving $R^2 = 0.992$ and RMSE = 6.71 using scikit-learn, pandas, and NumPy.
- Delivered a fully functional hackathon demo, demonstrating AI-driven decision support for e-commerce pricing strategy.

Additional Projects

- **Titanic Survival Prediction –** ML model (Decision Tree) predicting passenger survival (~79% accuracy).
- **HealthAI – Personalized Medical Recommendation System –** Web app predicting diseases and providing guidance; Flask deployment.
- **Travel & Tourism Recommendation System –** Flask app giving personalized travel suggestions using hybrid ML models.
- **Spam Message Classifier –** NLP-based spam detector (~99% accuracy) deployed with Streamlit.
- **E -Commerce Product Recommendation System –** Full-stack product recommendation using TF-IDF & Flask with real-time queries.

CERTIFICATES

- Machine Learning Specialization – Andrew Ng, Stanford Online | ID: 6J2J27TU3T2G
- Python for Data Science, AI & Development – IBM | ID: VFEX71365C0Q
- Collaborate Effectively for Professional Success – IBM | ID: MAYHO1EBOVJC
- Introduction to AI – Google | ID: H6RGE3R9RGKG
- AI/ML – Career Institute

HOBBIES

- Exploring AI/ML projects and technology trends
- Competitive programming and problem-solving
- Fitness and sports (gym, running